

MAT 1272 Statistics
Chapter 4: Basic Probability
Sections 4.1, 4.2, 4.3.1, 4.3.2, and 4.3.3

Learning Objectives

Goals for Today

By the end of this lesson, students should be able to:

- identify experiments, outcomes, and sample spaces;
- draw and read simple tree diagrams;
- distinguish simple events from compound events;
- compute classical probability;
- understand relative frequency and subjective probability;
- read marginal probabilities from a two-way table;
- compute conditional probabilities;
- distinguish mutually exclusive events from independent events;
- use the complement rule.

4.1 Experiments, Outcomes, and Sample Spaces

Experiment

An **experiment** is a process that produces one outcome.

Examples:

- Tossing a coin.
- Rolling a die.
- Selecting a student from a class.
- Selecting two workers from a company.

Outcome

An **outcome** is one possible result of an experiment.

One Toss of a Coin

For one toss of a coin, the possible outcomes are

$$H, \quad T.$$

Sample Space

The **sample space** is the set of all possible outcomes of an experiment.
It is usually denoted by

$$S.$$

Sample Spaces

For one toss of a coin:

$$S = \{H, T\}.$$

For one roll of a die:

$$S = \{1, 2, 3, 4, 5, 6\}.$$

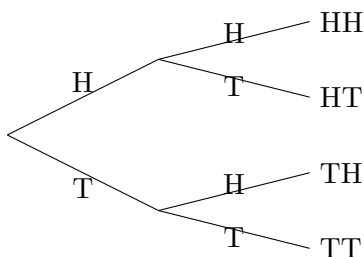
Tree Diagrams

Tree Diagram

A **tree diagram** is a visual way to list all possible outcomes of an experiment with stages.

Coin Tossed Twice

For two tosses of a coin, the tree diagram is:



The sample space is

$$S = \{HH, HT, TH, TT\}.$$

Simple and Compound Events

Event

An **event** is a collection of one or more outcomes of an experiment.

Simple Event

A **simple event** contains exactly one outcome.

Simple Event

For two coin tosses,

$$A = \{HH\}$$

is a simple event.

Compound Event

A **compound event** contains more than one outcome.

Compound Event

For two coin tosses, let B be the event that at least one head occurs.

$$B = \{HH, HT, TH\}.$$

This is a compound event.

4.2 Probability

Probability

Probability is a numerical measure of how likely an event is to occur.

Property 1

For any event A ,

$$0 \leq P(A) \leq 1.$$

A probability cannot be negative and cannot be greater than 1.

Property 2

The probabilities of all simple outcomes in a sample space add to 1.

$$\sum P(E_i) = 1.$$

Impossible and Certain Events

An impossible event has probability

0.

A certain event has probability

1.

Classical Probability

Classical Probability

Classical probability is used when all outcomes are equally likely.

Classical Probability Formula

If all outcomes are equally likely, then

$$P(A) = \frac{\text{number of outcomes favorable to } A}{\text{total number of outcomes}}.$$

One Toss of a Coin

For one toss of a fair coin,

$$S = \{H, T\}.$$

$$P(H) = \frac{1}{2} = 0.50.$$

$$P(T) = \frac{1}{2} = 0.50.$$

One Roll of a Die

Find the probability of rolling an even number.

The sample space is

$$S = \{1, 2, 3, 4, 5, 6\}.$$

The event is

$$A = \{2, 4, 6\}.$$

Thus,

$$P(A) = \frac{3}{6} = 0.50.$$

Relative Frequency Probability

Relative Frequency Probability

Relative frequency probability estimates probability using observed data.

Relative Frequency Formula

If an experiment is repeated n times and event A occurs f times, then

$$P(A) \approx \frac{f}{n}.$$

Defective Products

Suppose 10 defective products are found in a sample of 500 products.
Then

$$P(\text{defective}) \approx \frac{10}{500} = 0.02.$$

Law of Large Numbers

Law of Large Numbers

As the number of trials increases, the relative frequency of an event approaches the true probability of the event.

Coin Tossing

For a fair coin, the true probability of heads is

$$P(H) = 0.50.$$

If we toss a coin only a few times, the relative frequency of heads may not be close to 0.50.
If we toss the coin many times, the relative frequency of heads should get closer to 0.50.

Subjective Probability

Subjective Probability

Subjective probability is based on judgment, experience, information, or belief.

Examples of Subjective Probability

- The probability that a team wins next season.
- The probability that a student earns an A.
- The probability that a company succeeds with a new product.

4.3.1 Marginal Probability

Marginal Probability

A **marginal probability** is the probability of a single event without considering any other event.

Employee Opinion Table

Suppose 100 employees were asked whether they are in favor of or against paying high salaries to CEOs.

	In Favor	Against	Total
Male	15	45	60
Female	4	36	40
Total	19	81	100

Finding Marginal Probabilities

Use the row totals or column totals divided by the grand total.

Marginal Probabilities

$$P(\text{Male}) = \frac{60}{100} = 0.60.$$

$$P(\text{Female}) = \frac{40}{100} = 0.40.$$

$$P(\text{In Favor}) = \frac{19}{100} = 0.19.$$

$$P(\text{Against}) = \frac{81}{100} = 0.81.$$

Conditional Probability

Conditional Probability

Conditional probability is the probability that an event occurs given that another event has already occurred.

The notation

$$P(A \mid B)$$

is read as “the probability of A given B .”

Conditional Probability Formula

$$P(A \mid B) = \frac{\text{number of outcomes in both } A \text{ and } B}{\text{number of outcomes in } B}.$$

The event after the vertical bar is the condition and goes in the denominator.

Conditional Probability from the Table

Find

$$P(\text{In Favor} \mid \text{Male}).$$

Since we are given that the employee is male, we only look at the male row.

There are 60 males.

Of those 60 males, 15 are in favor.

Thus,

$$P(\text{In Favor} \mid \text{Male}) = \frac{15}{60} = 0.25.$$

Another Conditional Probability

Find

$$P(\text{Female} \mid \text{In Favor}).$$

Since we are given that the employee is in favor, we only look at the In Favor column.

There are 19 employees in favor.

Of those 19 employees, 4 are female.

Thus,

$$P(\text{Female} \mid \text{In Favor}) = \frac{4}{19} \approx 0.2105.$$

4.3.2 Mutually Exclusive Events

Mutually Exclusive Events

Two events are **mutually exclusive** if they cannot occur at the same time.

Coin Toss

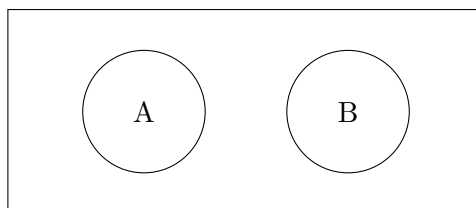
When tossing one coin, the events

$$H = \{\text{Head}\}$$

and

$$T = \{\text{Tail}\}$$

are mutually exclusive because we cannot get both head and tail on one toss.



Mutually exclusive: no overlap

Die Example

Let

$$A = \{2, 4, 6\}$$

be the event that an even number is rolled.

Let

$$B = \{1, 3, 5\}$$

be the event that an odd number is rolled.

These events are mutually exclusive.

Not Mutually Exclusive

Let

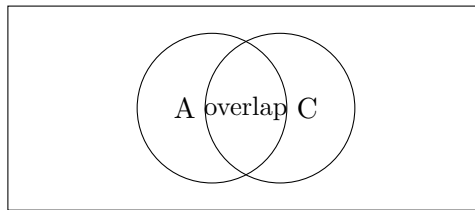
$$A = \{2, 4, 6\}$$

and

$$C = \{1, 2, 3, 4\}.$$

These events are not mutually exclusive because they share outcomes:

$$A \cap C = \{2, 4\}.$$



Not mutually exclusive: overlap exists

4.3.3 Independent and Dependent Events

Independent Events

Two events are independent if the occurrence of one event does not change the probability of the other event.

Independence Rule

Events A and B are independent if

$$P(A \mid B) = P(A)$$

or equivalently

$$P(B \mid A) = P(B).$$

Coin Tosses

For two tosses of a fair coin, the result of the first toss does not affect the result of the second toss.

Therefore, the two tosses are independent.

Dependent Events

Two events are dependent if the occurrence of one event changes the probability of the other event.

Dependent Events

Suppose we draw a card from a deck and do not replace it.

The probability of the second draw depends on what happened on the first draw.

Therefore, the events are dependent.

Important Distinction

Mutually exclusive and independent are not the same idea.

- Mutually exclusive means events cannot happen together.
- Independent means one event does not affect the probability of the other.

Complementary Events

Complementary Events

The complement of event A , denoted A^c , is the event that A does not occur.

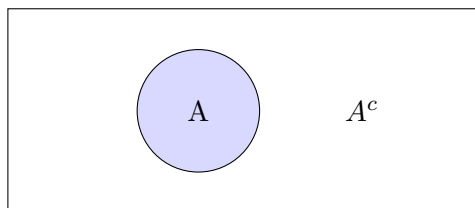
Die Example

Let

$$A = \{\text{even number}\} = \{2, 4, 6\}.$$

Then

$$A^c = \{\text{not even}\} = \{1, 3, 5\}.$$



The complement is everything outside A

Complement Rule

Also,

$$P(A^c) = 1 - P(A).$$
$$P(A) + P(A^c) = 1.$$

Complement Example

Suppose

$$P(A) = 0.65.$$

Then

$$P(A^c) = 1 - 0.65 = 0.35.$$

Group Activity

Work in Groups

A die is rolled once.
Let

$$A = \{2, 4, 6\}$$

and

$$B = \{4, 5, 6\}.$$

Answer the following:

1. Write the sample space.
2. Find $P(A)$.
3. Find $P(B)$.
4. Find $A \cap B$.
5. Are A and B mutually exclusive?
6. Find A^c .
7. Find $P(A^c)$.

Second Group Activity

Two-Way Table Practice

Use the table below.

	In Favor	Against	Total
Male	15	45	60
Female	4	36	40
Total	19	81	100

Find:

1. $P(\text{Female})$
2. $P(\text{Against})$
3. $P(\text{In Favor} \mid \text{Female})$
4. $P(\text{Female} \mid \text{In Favor})$
5. Are “Female” and “In Favor” independent? Explain briefly.

Summary

Chapter 4 Key Ideas

- The sample space S contains all possible outcomes.
- A simple event has one outcome.
- A compound event has more than one outcome.
- Classical probability:

$$P(A) = \frac{\text{favorable outcomes}}{\text{total outcomes}}.$$

- Relative frequency probability:

$$P(A) \approx \frac{f}{n}.$$

- Conditional probability:

$$P(A \mid B) = \frac{\text{outcomes in both } A \text{ and } B}{\text{outcomes in } B}.$$

- Mutually exclusive events cannot occur together.
- Independent events do not affect each other's probabilities.
- Complement rule:

$$P(A^c) = 1 - P(A).$$